

A note on Γ -supercyclicity of dissipative composition operators

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Abstract

We record an affirmative answer to the question on arbitrary- Γ supercyclicity posed in Oberwolfach Report 19/2024. In the report's separable L^p setting, a dissipative composition operator of bounded distortion is Γ -supercyclic if and only if its associated bilateral weighted shift is Γ -supercyclic, for every subset $\Gamma \subseteq \mathbb{C}$. The observation follows from existing results of Abbar and of Abbar–Kuznetsova, together with a sequence-space representation of the composition operator. We give a short direct proof to make the application explicit. Neither the scalar criterion nor the amplification mechanism is claimed as new.

1 The question and the statement

For a bounded operator T on a complex Banach space and a set $\Gamma \subseteq \mathbb{C}$, call T Γ -*supercyclic* if some vector x has dense orbit

$$\{\lambda T^n x : n \geq 0, \lambda \in \Gamma\}.$$

This terminology follows Charpentier, Ernst, and Menet [3]. In her contribution to Oberwolfach Report 19/2024, D'Aniello, reporting joint work with Maiuriello, asks whether the equivalence of this property for a dissipative composition operator of bounded distortion and its associated shift holds for arbitrary Γ [6, p. 1083]. The cases of ordinary hypercyclicity and real or complex supercyclicity are part of the preceding work [4, 5].

We use the standing separability convention in that report. Let $(X, \mathcal{B}, \mu)$ be a σ -finite measure space with separable complex $L^p(X, \mu)$, where $1 \leq p < \infty$. Let $f : X \rightarrow X$ be a bijective bimeasurable transformation such that, for finite constants A, B ,

$$\mu(f^{-1}C) \leq A\mu(C), \quad \mu(fC) \leq B\mu(C) \quad (C \in \mathcal{B}). \quad (1)$$

Thus $T_f\varphi = \varphi \circ f$ is bounded and invertible. Suppose a measurable wandering set W satisfies

$$X = \dot{\bigcup}_{j \in \mathbb{Z}} f^j W, \quad 0 < \mu(W) < \infty,$$

where a null remainder can be discarded. Write $m_j = \mu(f^j W)$. Bounded distortion means that some $K \geq 1$ satisfies

$$K^{-1}m_j\mu(C) \leq \mu(f^j C)m_0 \leq Km_j\mu(C) \quad (j \in \mathbb{Z}, C \subseteq W \text{ measurable}). \quad (2)$$

These are the dissipative hypotheses recalled in [6, pp. 1079–1080].

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Theorem 1. *Under these assumptions, let B_w on $\ell^p(\mathbb{Z})$ be given by*

$$(B_w a)_j = w_{j+1} a_{j+1}, \quad w_j = \left(\frac{m_{j-1}}{m_j} \right)^{1/p}.$$

For every $\Gamma \subseteq \mathbb{C}$, T_f is Γ -supercyclic if and only if B_w is Γ -supercyclic.

The contribution of this note is the explicit application to the reported question. Abbar's scalar criterion [1, Theorem A] and his Γ -supercyclicity criterion [1, Theorem 2] already imply the amplification step below. Theorem B of Abbar–Kuznetsova [2] gives a second route, described in Section 4. We reproduce the elementary estimates for ease of verification, not as additional novelty claims.

2 The scalar criterion and its amplification

Let $c_j > 0$ and suppose the translation $(Sa)_j = a_{j+1}$ is bounded on

$$Y = \ell^p(\mathbb{Z}, c), \quad \|a\|_c^p = \sum_{j \in \mathbb{Z}} c_j |a_j|^p.$$

For a nonzero separable complex Banach space E , use the same translation S_E on $Y_E = \ell^p(\mathbb{Z}, c; E)$, with norm $\|u\|_c^p = \sum_j c_j \|u_j\|_E^p$. Boundedness follows from $\sup_j c_{j-1}/c_j < \infty$. Put $\Gamma^* = \Gamma \setminus \{0\}$.

Lemma 2 (Finite-block form of Abbar's criterion). *The following condition is necessary for S to be Γ -supercyclic, and sufficient for S_E to be Γ -supercyclic: for every finite nonempty $F \subset \mathbb{Z}$, $\eta > 0$, and $L \geq 0$, there are $n > L$ and $\lambda \in \Gamma^*$ with*

$$\frac{(\sum_{j \in F} c_{j+n})^{1/p}}{|\lambda|} < \eta, \quad |\lambda| (\sum_{j \in F} c_{j-n})^{1/p} < \eta. \quad (3)$$

For scalar coefficients, this is the finite-block formulation of [1, Theorem A], with weight $\omega_j = c_j^{1/p}$. To pass from that theorem's single-offset condition to a finite block F , choose an integer $q \geq \max(F \cup \{1\})$ and use

$$c_{j \pm n}^{1/p} \leq \|S\|^{q-j} c_{q \pm n}^{1/p} \quad (j \in F).$$

Finite sums and finite maxima then give equivalent conditions. The sufficiency argument below is the usual universality-criterion argument underlying [1, Theorem 2].

Proof. Suppose $x \in Y$ has a dense Γ -orbit. Every nonempty open set is reached at arbitrarily large times: for each N , the closed set $\bigcup_{n=0}^N \mathbb{C} S^n x$ has empty interior in the infinite-dimensional space Y . Also the support of x is unbounded above, since otherwise every forward iterate would vanish above a common index.

Fix F and choose $k \geq \max F$ with $x_k \neq 0$. With e_j denoting the coordinate vectors, set $v = \sum_{j \in F} e_j$. There are $n_r \rightarrow \infty$ and $\lambda_r \in \Gamma^*$ such that $\delta_r = \|\lambda_r S^{n_r} x - v\|_c \rightarrow 0$. For large r , each $j \in F$ satisfies $|\lambda_r x_{j+n_r}| > 1/2$, whence

$$\frac{(\sum_{j \in F} c_{j+n_r})^{1/p}}{|\lambda_r|} \leq 2 \left(\sum_{j \in F} c_{j+n_r} |x_{j+n_r}|^p \right)^{1/p} \rightarrow 0.$$

The limit follows because $F + n_r$ escapes every finite interval and $x \in Y$. For large r , $k - n_r \notin F$, so

$$|\lambda_r| c_{k-n_r}^{1/p} |x_k| \leq \delta_r.$$

Writing $M = \|S\|$, the coordinate estimate $c_{t-d}^{1/p} \leq M^d c_t^{1/p}$ for $d \geq 0$ yields

$$|\lambda_r| \left(\sum_{j \in F} c_{j-n_r} \right)^{1/p} \leq \left(\sum_{j \in F} M^{p(k-j)} \right)^{1/p} \frac{\delta_r}{|x_k|} \longrightarrow 0.$$

This proves necessity, with the same allowed scalar in both estimates.

Conversely, assume (3). Let $U, V \subset Y_E$ be nonempty open sets. Choose finitely supported $a \in U$ and $b \in V$. Let a finite nonempty F contain both supports, and bound their coordinate norms by $H \geq 1$. For $n \geq 0$, define the right translation of a finite-support sequence by $(R_n b)_j = b_{j-n}$, so $S_E^n R_n b = b$. The condition supplies n, λ for which

$$z = a + \lambda^{-1} R_n b \in U, \quad \lambda S_E^n z = \lambda S_E^n a + b \in V :$$

indeed, the two errors are bounded by H times the first and second quantities in (3), respectively, and can be arbitrarily small.

Take a countable base (V_q) of Y_E . For each q , the set

$$G_q = \bigcup_{n \geq 0, \lambda \in \Gamma} \{z : \lambda S_E^n z \in V_q\}$$

is open by continuity and dense by the preceding argument. Baire's theorem makes $\bigcap_q G_q$ dense and nonempty; its elements are Γ -supercyclic vectors. No countability or regularity assumption on Γ is needed. $\square$

Proposition 3 (Amplification). *The operators S and S_E are Γ -supercyclic for exactly the same subsets $\Gamma \subseteq \mathbb{C}$.*

Proof. The forward implication is Lemma 2. For the reverse, choose $e \in E$ and $\ell \in E^*$ with $\ell(e) = 1$. The bounded surjection $Q : Y_E \rightarrow Y$, $(Qu)_j = \ell(u_j)$, satisfies $QS_E = SQ$ and sends a dense Γ -orbit to a dense Γ -orbit. If $\Gamma^* = \emptyset$, neither operator has such an orbit, so this case also satisfies the equivalence. $\square$

3 The composition model and the answer

Proof of Theorem 1. Set $c_j = m_j/m_0$ and $E = L^p(W, \mu|_W)$. The measure bounds (1) ensure $0 < m_j < \infty$ and boundedness of both scalar translation directions. The space E is nonzero and separable: extension by zero embeds it as a closed subspace of $L^p(X, \mu)$.

For measurable $C \subseteq W$, define $\nu_j(C) = \mu(f^j C)$. Bimeasurability and injectivity make this a measure, and bounded distortion says

$$K^{-1} c_j \mu|_W \leq \nu_j \leq K c_j \mu|_W.$$

Define $J\varphi = (\varphi \circ f^j|_W)_{j \in \mathbb{Z}}$. The measure-class bounds ensure that J is well-defined on equivalence classes. Decomposing the integral over the wandering partition gives

$$\|\varphi\|_p^p = \sum_j \int_W |\varphi \circ f^j|^p d\nu_j, \quad K^{-1} \|J\varphi\|_c^p \leq \|\varphi\|_p^p \leq K \|J\varphi\|_c^p. \quad (4)$$

For $u \in Y_E$, choose measurable representatives and define φ on $f^j W$ by $\varphi = u_j \circ f^{-j}$. This countable gluing is measurable and (4) shows that $\varphi \in L^p(X, \mu)$. Thus J is a bounded linear isomorphism onto Y_E , and direct substitution gives $JT_f = S_E J$.

Finally, $D : \ell^p(\mathbb{Z}) \rightarrow Y$, $(Da)_j = c_j^{-1/p} a_j$, is an onto isometry, with

$$(DB_w a)_j = c_j^{-1/p} w_{j+1} a_{j+1} = c_{j+1}^{-1/p} a_{j+1} = (SDa)_j.$$

Complex-linear conjugacies preserve Γ -orbits for every fixed Γ . Proposition 3 therefore gives $T_f \Gamma$ -supercyclic $\iff S_E \Gamma$ -supercyclic $\iff S \Gamma$ -supercyclic $\iff B_w \Gamma$ -supercyclic. $\square$

Separability cannot be dropped from this application merely because the measure is σ -finite. Every projective orbit is contained in the separable closed span of a countable orbit. For example, on $X = \mathbb{Z} \times \Omega$, let $\Omega = \{0, 1\}^I$, with I uncountable, carry the product of fair Bernoulli probability measures, give layer j mass $2^{-|j|}$ times that measure, and set $f(j, t) = (j+1, t)$. Bounded distortion holds with $K = 1$ and T_f is bounded and invertible. Its L^p space is nonseparable, while the scalar shift is hypercyclic by (3). This is a scope clarification, not an additional novelty claim.

4 A direct deduction from the translation theorem

For completeness, the following deduction identifies the relation to Abbar–Kuznetsova [2, Theorem B] without using the proof of Lemma 2. On the discrete group $\mathbb{Z}$, take the translation parameters $\{-n : n \geq 0\}$ and weight $\omega(j) = c_j^{1/p}$. Their translation convention gives $T_{-n}a(j) = a(j+n)$. The weight is admissible because S is bounded. Counting measure and an exceptional-set tolerance smaller than one force the retained set in their criterion to be the whole finite set. Scalar Γ -supercyclicity therefore supplies the two maximum versions of (3). The times can exceed any fixed L : over finitely many times $n \leq L$, the product of the two positive maxima has a positive minimum, independently of λ .

Now use $\mathbb{Z}^2$, parameters $\{(-n, 0) : n \geq 0\}$, and weight $\omega(j, k) = c_j^{1/p}$. A finite set has a finite first-coordinate projection, so the same estimates verify their criterion. This gives Γ -supercyclicity on $\ell^p(\mathbb{Z}, c; \ell^p(\mathbb{Z}))$. For arbitrary separable E , choose unit vectors e_k with dense linear span. The map

$$R : \ell^p(\mathbb{Z}) \rightarrow E, \quad R(a) = \sum_{k=1}^{\infty} 2^{-k} a_k e_k$$

is bounded, since $\|R(a)\| \leq \|a\|_{\infty} \leq \|a\|_p$, and has dense range. Its coordinatewise extension has dense range and commutes with translation, hence transfers a dense Γ -orbit. The functional quotient gives the converse. This proves Proposition 3 as a consequence of the earlier theorem. The case $\Gamma^* = \emptyset$ is immediate separately.

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